Portfolio: Business Process Modelling and Enterprise Architecture

Hospital Patient Referral, Treatment and Administration System

Module: UFCEP4-0-3 | **Assessment:** Portfolio (50%) | **School of Computing and Creative Technologies, UWE Bristol** | **Submission:** 22 October 2026

Group: max (BPMN and integration), finn (external workers, i* modelling), henry (requirements and traceability, alignment evaluation, agile and documentation), peter (forms, QA and testing). Four members, agreed with the module tutor.

Contents

Section	Deliverable	Criterion
1	Strategic BPMN models and i* SD/SR modelling rationale (Tasks 01 and 02)	20% + 20%
2	Evaluation of alignment with requirements (Task 03)	15%
3	Evaluation of acceptance criteria (Task 04)	15%
4	Agile delivery record: Sprints 3 and 4, standups, reviews, retrospectives, feedback actions	30%
References	Sources used for the notation, the framework and the case study	—
Appendix	Model inventory, validation, deployment and execution evidence, defect log	—

Files submitted

Deliverable	Format
Operational model `01_Operational_Merged`	`.bpmn` + PDF
Strategic models `S1_Strategic_Landscape`, `S2_Strategic_Referral_To_Authorisation`, `S3_Strategic_Treatment_To_Aftercare`	`.bpmn` + PDF
Camunda Forms for every user task	68 `.form` files
i* models: SD and SR	PDF + editable SVG and JSON source
Alignment evaluation, acceptance evaluation, agile record	this document

A note on the evidence

Four things are kept separate throughout. The **models** state intended behaviour and were validated with the Camunda Modeler toolchain. The **deployed system** is the same model set deployed to a running Camunda 8 cluster and executed: eighteen acceptance tests were run against it and all eighteen passed. The **initial release** covers referral through to treatment scheduling; the correspondence, enquiry, follow-up and reporting scope is modelled but not implemented. The **Sprint 1 and Sprint 2 working files** are earlier increments whose defects are catalogued in the appendices.

Deploying the models found four defects that model validation had not: three BPMN XSD sequence violations and one runtime form-binding defect. They are recorded, fixed and re-verified in section 5.

Task 01 and Task 02: BPMN models and i* socio-technical models

Part A — Task 01: Strategic BPMN process models (LO1, LO3)

A.1 Scope and abstraction

Task 01 asks for high-level BPMN models of the initial release that show the broader business processes, the organisational responsibilities and the external interactions, and that explain how the strategic activities relate to the operational model. Because no single readable diagram can do all of that, the view is presented at two levels.

Model	File	Abstraction
O1 operational merged model	`O1_Operational_Merged.bpmn`	Level 3 — detailed operational tasks, 8 lanes, 59 nodes, 8 external participants
S1 strategic process landscape	`S1_Strategic_Landscape.bpmn`	Level 1 — one hospital pool, no internal roles, 24 nodes, 7 external participants
S2 strategic collaboration, referral to authorisation	`S2_Strategic_Referral_To_Authorisation.bpmn`	Level 2 — 6 lanes, 36 nodes
S3 strategic collaboration, treatment to aftercare	`S3_Strategic_Treatment_To_Aftercare.bpmn`	Level 2 — three processes (7, 3 and 1 lanes), 74 nodes

What each model is for, concretely. S1 answers "what is the system boundary and who does the hospital exchange messages with". It is the only view in which the seven external participants and the fifteen message flows can be seen together, and it is the view a new stakeholder should read first. S2 answers "who is accountable from referral to an authorised treatment request", which is the half of the pathway where administrative error is most damaging because clinical decisions are adjacent to administrative ones. S3 answers the same question for the half where money changes hands, and adds the three areas the case study describes but the first two models do not reach: correspondence and its escalation ladder, patient enquiries, and reporting. O1 answers "what does the initial release actually do, step by step".

The split between S2 and S3 is not arbitrary. It falls at the point where the Treatment Booking Request is authorised and passes from clinical responsibility into administrative and financial responsibility — which is the boundary the case study is most insistent about, and therefore the most useful place to break a diagram.

Why these levels. Level 1 is deliberately a value chain: one box per business phase, no roles, no decisions internal to a team. It exists so that the system boundary and the external dependencies can be read on one page, and so that the wider business process can be compared with the implemented solution. Level 2 keeps the same boundary and external participants but attributes every activity to the unit accountable for it. Level 3 is the operational detail.

Level 2 is the right stopping point because the case study's central risk is not a missing screen but **confused responsibility**: administrative staff must not make clinical judgements, clinical staff must not approve refunds without financial authority, and treatment changes made informally must not be processed. Those are properties of the division of labour and are only visible when activities are attributed to roles. Each level 2 activity therefore carries a `documentation` element naming the operational step it decomposes into.

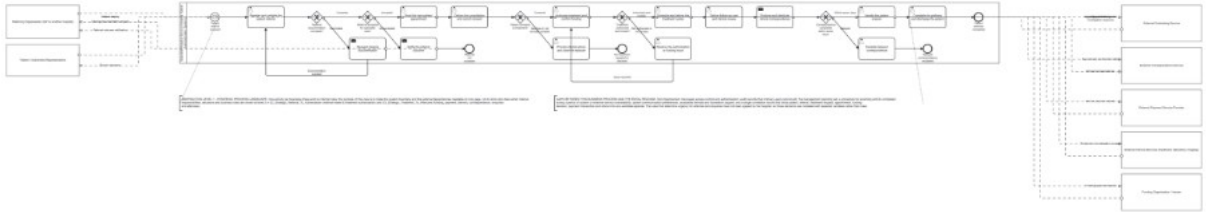


Figure 1. S1 level 1 landscape: one hospital pool, one box per phase, every external participant as a collapsed black-box pool.

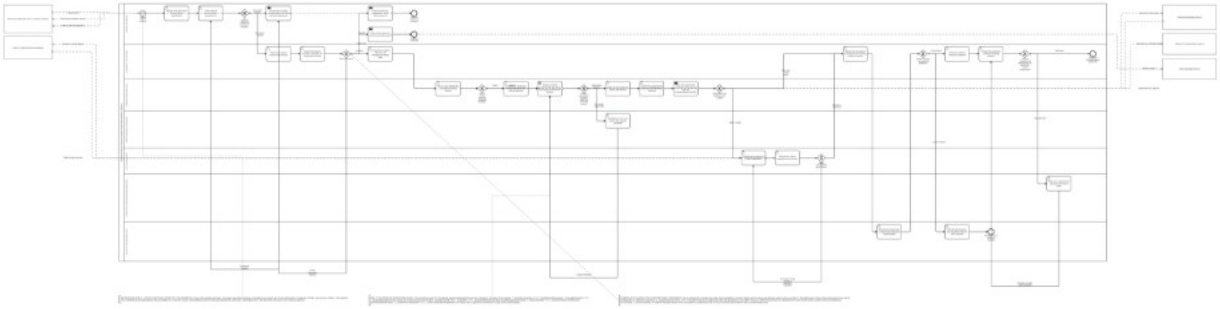


Figure 2. S2 level 2 collaboration: referral intake and triage, appointment booking, consultation and consent, clinical authorisation. Readable PDF: `02_BPMN_Models/readable_exports/S2_Strategic_Referral_To_Authorisation.pdf`.

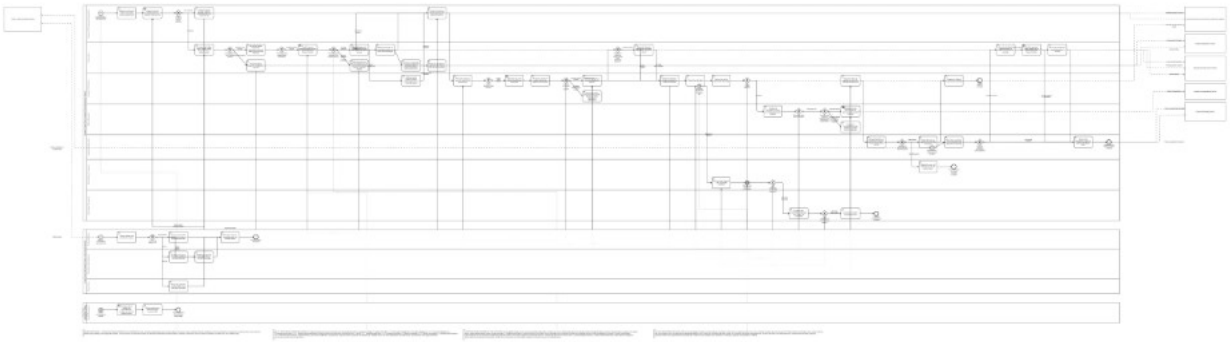


Figure 3. S3 level 2 collaboration: funding and payment, treatment delivery and review cycles, authorised modification, correspondence and escalation, follow-up and aftercare, patient enquiries, management reporting. Readable PDF: `02_BPMN_Models/readable_exports/S3_Strategic_Treatment_To_Aftercare.pdf`.

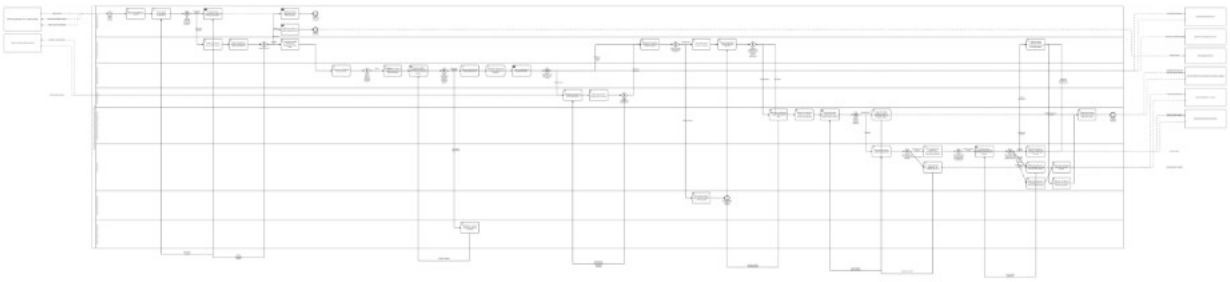


Figure 4. O1 level 3 operational model: the merged operational process from referral to a confirmed treatment schedule. Readable PDF: `02_BPMN_Models/readable_exports/O1_Operational_Merged.pdf`.

A.2 Notation and conventions

White-box participants. S2 declares one hospital pool with seven lanes (Medical Secretaries; Consultant / Clinical Team; Outpatient Bookings Team; Patient Pathway Coordinators; Call Handling Team; Treatment and Chemotherapy Bookings Team; Clinical Nurse Specialist Team). S3 declares a main hospital pool with seven lanes and **two further processes**: patient enquiry handling (three lanes) and management reporting (one lane). Only lanes that carry work are declared — the original Sprint 1 and 2 files declared lanes for the Finance Team, Clinical Nurse Specialist Team, Call Handling Team, Administrative Management and Outpatient Bookings Team and then left them empty, which asserts a responsibility the process does not discharge.

Enquiry handling and reporting are separate processes rather than lanes because neither is a step in an individual patient's pathway: a patient may telephone at any time, and reporting runs on a monthly cycle.

The models are authored in BPMN 2.0 (Object Management Group, 2013) and carry the Camunda 8 execution attributes — form references, job types, message subscriptions and timer definitions — described in the platform documentation (Camunda, 2025). Every element is drawn from the BPMN 2.0 meta-model; no proprietary notation is introduced.

Black-box participants. External and upstream organisations are collapsed pools: the referring organisation, the patient or authorised representative, the external scheduling service, the external correspondence service, the external payment service provider, external clinical services, the funding organisation or insurer, and another specialist service that can receive a redirected referral. They are collapsed because the case study does not describe their internal behaviour and the hospital has no authority over it.

External interactions. Every exchange with the outside world is an explicit labelled message flow, including the ones that are easy to omit: the missing-documentation request, the referral outcome notification, the redirect, slot queries and responses, appointment letters, telephone contact attempts and their outcome, funding approval requests and responses, secure payment requests and payment status, refund requests, blood test requests, clinic letter distribution, follow-up requests, and the cancellation or non-attendance notice.

Implementation-facing detail. User tasks carry a Camunda form reference and a `candidateGroups` value naming the owning team; automated activities carry a Zeebe job type (for example `query-available-appointment-slots`, `request-secure-payment`, `request-refund`, `distribute-clinical-correspondence`). This is why the models pass the Camunda 8 compatibility linter with zero reports: an unassigned user task or a service task without a task definition is a real defect in a Camunda 8 model.

A.3 Merging the three Sprint 1 and Sprint 2 fragments

The group produced three BPMN files in Sprints 1 and 2. They were written at different times, by different owners, against different execution platforms, and each covered only part of the pathway. Merging them required correcting the defects the merge exposed. The audit is reproducible: it parses each file with the same moddle schema Camunda Modeler uses on open and runs the Camunda Modeler lint engine headlessly.

Defect	Correction
Mixed execution platforms. Two files use Zeebe; the third uses Camunda 7 (`camunda:type="external"`, `${}` expressions) and declares no execution platform	All models targeted at Camunda 8 (Zeebe) 8.10; expressions converted to FEEL; job workers declared with `zeebe:taskDefinition`
Duplicate element identifiers across files (`Participant_0xulon7`, `Process_0fl9bxn` and others reused with different meanings)	One naming convention: `P_`, `L_`, `T_`, `G_`, `E_`, `F_`, `MF_`, `Msg_`
A pool labelled "External Payment Service Provider" pointed at the hospital's own process; two pools shared that name, and two shared "External Scheduling Service"	One participant per organisation, each name matched to its own process

Defect	Correction
Empty processes and empty lanes: 7 empty processes and 5 empty lanes in one file, 6 empty black-box participants in another, 6 empty lanes in the first	Only lanes and participants that carry work are declared
Dangling and unreachable elements: "Record missing document list" had no outgoing flow, so a referral with missing documents could never continue; the rejection and redirect notices had no incoming flow; another file's payment task had neither	The missing-documentation request returns to the completeness check; rejection and redirect terminate in named end events
A rejected or redirected referral was forwarded to the booking team	The decision gateway has four labelled branches; only the accepted branch reaches the Outpatient Bookings Team
Five exclusive gateways with no default flow, which fail at runtime with "no outgoing sequence flow could be selected"	Every diverging exclusive gateway declares a default flow; every non-default branch carries a condition
Inconsistent decision values: `"reject"`, `"redirect"`, `"Request more info"` and `"Accept"` for the same variable	One controlled value set used consistently
Roles in the wrong lanes: Medical Secretaries held slot booking; the telephone task sat in the Call Handling lane but was assigned to the bookings team	Work attributed to the team the case study makes accountable
Non-canonical XML: no `incoming` or `outgoing` elements on any flow node; task definitions on end events; an unknown `camunda:gatewayDirection` attribute	Canonical serialisation; outbound messages are send tasks; unknown attributes removed
The 14-day telephone rule was missing, contact attempts were not recorded, and a 14-day timer that does not exist in the case study terminated the process	The 14-day rule is an explicit gateway; attempts and outcomes are recorded
No coverage of refunds, non-attendance, delayed-letter escalation, enquiries, treatment modification or follow-up	Modelled in S3; where not implemented, recorded as a gap

A.4 Validation, deployment and execution

The delivered models were checked five ways.

- **Schema round-trip** with `bpmn-moddle` 10.1.0 using the Zeebe 1.18 and modeler extensions — the same parse that happens when a file is opened in Camunda Modeler. Result: **zero warnings** in all four models.
- **Camunda compatibility lint** with `@camunda/linting` 3.57.0 configured for `Camunda Cloud` diagrams, the library behind the Modeler "Problems" panel. Result: **zero reports** in all four models.
- **Structural audit**: connectivity recomputed from the sequence flows, reachability from start and boundary events, paths to an end event, gateway default flows, duplicate identifiers and participant names, empty lanes and empty processes. Result: **no defects**.
- **Deployment** to a running Camunda 8 Run 8.10.0-alpha5 cluster through the Orchestration Cluster API, together with 68 Camunda Forms generated from the models. Result: **all six process definitions deploy**.
- **Execution**: eighteen acceptance test cases driven against the deployed operational model. Result: **18 of 18 passed**, with the process instance key and full element-instance trace retained for each.

Deployment and execution found four defects that checks 1 to 3 could not:

ID	Defect	How it surfaced
DEP-01	`bpmn:documentation` emitted as a child of `bpmn:definitions`, which the BPMN 2.0 XSD does not permit	Deployment rejected: `cvc-complex-type.2.4.a`
DEP-02	Event definitions serialised before `incoming`/`outgoing`; the XSD requires them after	Deployment rejected: `cvc-complex-type.2.4.a`
DEP-03	Text annotations and associations serialised before the last sequence flow	Deployment rejected: `cvc-complex-type.2.4.a`
DEP-04	Every user task raised FORM_NOT_FOUND at runtime because the referenced forms were not deployed	Live execution: incident on the first user task of every instance

The first three are exactly what a lenient parser hides: the models passed the moddle round-trip and the linter with zero findings and rendered correctly in the Modeler, yet the engine's XSD validator rejected them. DEP-04 only appears once an instance is started. All four are fixed at source and closed by the clean execution run; the full log is in `08_Repository_Documents/defect-log.md`.

A.5 How strategic activities relate to the operational model

Three devices carry the link: the `documentation` element on every strategic activity naming the operational step it decomposes into; the form and job-type attributes that identify the concrete mechanism; and the task identifiers, which are stable and are the trace target in Task 03.

Strategic activity	Operational realisation (O1)	Verified by
Check referral documentation completeness	`T_CheckReferralDocuments`	TC-01, TC-02
Request the missing documentation	`T_RequestMissingDocuments` (job `request-missing-documentation`)	TC-02
Perform the clinical review; record decision and clinician	`T_ClinicalReviewReferral`, `T_RecordReferralDecision`	TC-01, TC-03, TC-04
Request available appointment slots; resolve capacity	`T_QueryAvailableSlots`, `T_ResolveCapacity`	TC-05
Telephone the patient within 14 days; record the attempt	`T_TelephonePatient`, `T_RecordContactAttempt`	TC-06, TC-07, TC-08
Record consent; create and authorise the Request	`T_RecordPatientConsent`, `T_CreateTreatmentBookingRequest`	TC-09, TC-10, TC-13
Request treatment capacity; hold pending	`T_RequestTreatmentCapacity`, `T_KeepPending`	TC-11
Determine funding; record approval	`T_DetermineFundingRoute`, `T_RecordFundingApproval`	TC-12, TC-13
Send the secure payment request; record the result	`T_SendSecurePaymentRequest`, `T_RecordPayment`	TC-14
Handle a declined payment; investigate an unconfirmed one	`T_HandlePaymentFailure`, `T_MarkForInvestigation`	TC-15, TC-16
Authorise urgent treatment without confirmed payment	`T_AuthoriseUrgentTreatment`, `T_ReferUnconfirmedPayment`	TC-17
Treatment modification, clinic letters, escalation, refunds, enquiries, reporting	**not implemented** in the initial release	gap

A.6 Gaps between the business process and the implemented solution

Role-based access control, authentication and immutable audit records. Stated as mandatory in the case study but not implemented as process behaviour, because they are cross-cutting system services. The models show the actions that must be audited but not the authorisation mechanism. Recorded as *not supported* in Task 03.

Management reporting. The reporting process is modelled and deployed with a monthly timer, but no report has been produced. Recorded as *out of scope for the initial release*.

Availability and downtime. The models assume the external services respond; the requirement to record activity during an outage is not modelled because the fallback procedure had not been agreed.

Urgency rules. The gateways exist but the hospital has not agreed the rules that decide them, so urgency is modelled with asserted variables and the ambiguity is documented.

Communication preferences and accessible formats. Modelled in S3 but not implemented; the initial release does not capture the preference.

Episode-level traceability. The components of the auditable relationship between patient, referral, request, appointment, funding decision, payment and refund are recorded, but no single correlation record binds them into one episode.

Part B — Task 02: Socio-technical modelling using i* (LO2, LO3)

B.1 Actors and the position of the models

The i* models were developed in Sprints 3 and 4, **after** the initial release was built. They did not inform the original design and are used as an evaluation lens, not a design input.

Actors were identified by asking, for each sentence of the case study, who is accountable for the outcome described. That produced nine intentional human actors inside the hospital and seven external actors, plus one deliberate omission: the HPAS itself is **not** an actor, because it is a system rather than an agent and has no goals of its own. The Clinical Nurse Specialist Administrative Support Team is subsumed into the Clinical Nurse Specialist Team because the case study attributes no distinct goal, task or resource to it.

Both diagrams use i* notation: actors and their intentional elements, means-end and task-decomposition links, and the four dependency types (goal, soft goal, task and resource), as defined by Yu (1995) and specified in the iStar 2.0 language guide (Dalpiaz, Franch and Horkoff, 2016). The dependency type is chosen from the case study wording rather than by default; section B.4 records where the wording was ambiguous.

B.2 The Strategic Dependency model

The SD model contains **16 actors and 40 dependencies**, each typed as a goal, soft goal, task or resource.

Resource dependencies dominate, which is the socio-technical insight the process models conceal. The hospital depends on the referring organisation for complete referral documentation, on the Consultant for a decision with a recorded reason, on the external scheduling service for available slots, on external clinical services for treatment, laboratory and imaging capacity, on the funding organisation for an authorisation with an amount and its limitations, on the payment provider for the payment status, and on the patient for informed consent and accurate identification. A resource dependency means the depender needs the *thing*, not the *activity*, and cannot control how it is produced. That is the root of several case-study failure modes: the hospital cannot make the scheduling service have a slot, and cannot make an insurer approve funding.

Task dependencies show delegated work: the Consultant depends on the Medical Secretaries to administer letters and on the Patient Pathway Coordinators to chase them; the Finance Team depends on the payment provider to process a payment and return the status; the Call Handling Team depends on the Clinical Nurse Specialist Team for clinical enquiries and on the Finance Team for payment enquiries.

Goal dependencies cannot be satisfied by performing a task correctly — they require the dependee to *achieve* something. The patient depends on the Consultant for *receive clinically appropriate treatment*, and the Treatment Bookings Team depends on the Finance Team for *funding confirmed before treatment*.

Soft goal dependencies carry expectations that are judged rather than verified, and which conflict: the recording management needs in order to demonstrate accountability is the same recording that reduces clinical time. Four are modelled — the patient's expectation that personal and financial information is protected, the Consultant's expectation that administrative burden is minimised, management's expectation that pathway delays are visible, and the patient's expectation that treatment is delivered without avoidable delay. None can be marked satisfied; all four can be helped or hurt by design choices.

The choice of dependency type is not cosmetic, and one example shows why. When the Outpatient Bookings Team asks the external scheduling service for an appointment, that is a **resource** dependency: the team needs a slot to exist. When the same team asks the Consultant to confirm the clinical priority of a referral, that is also a resource dependency — the team needs the priority as an input. But when the Treatment Bookings Team needs funding to be in place before treatment, that is a **goal** dependency: the dependee must *achieve* something, and no amount of correct task performance by the bookings team substitutes for it. Modelling that distinction is what allows Task 03 to say that some requirements are only partly supported even though every corresponding activity exists in the process.

Dependency count. Forty dependencies across sixteen actors is a deliberately bounded model. Adding a dependency for every exchange in the process models would produce a diagram with no analytic value; the criterion applied was whether the exchange represents something one actor needs from another that it cannot produce itself.

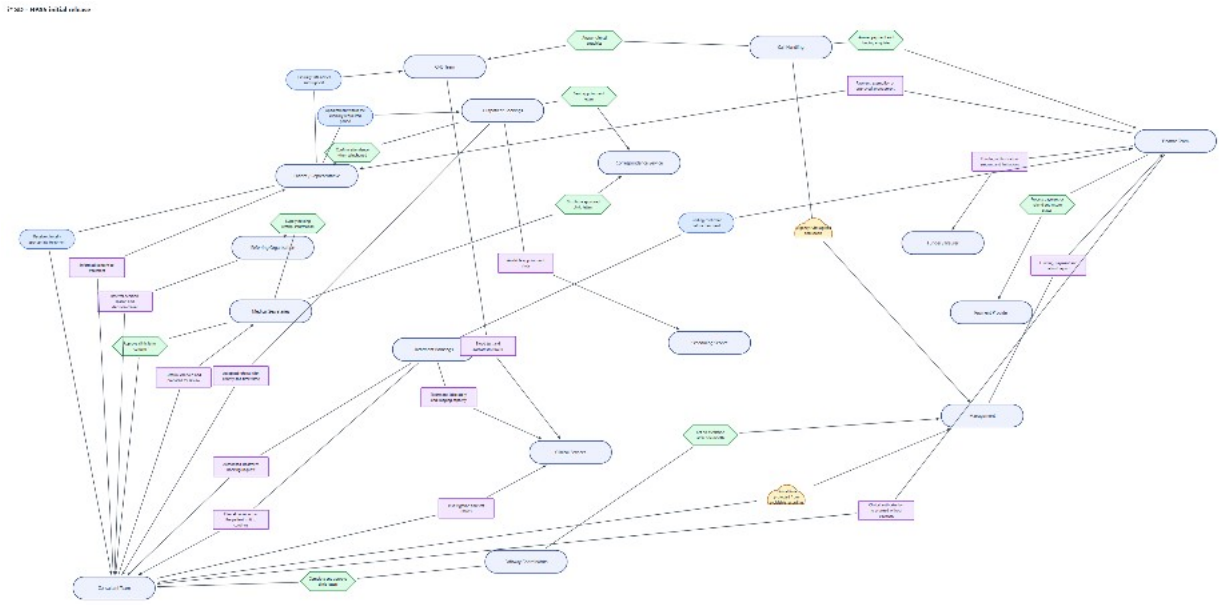


Figure 5. i SD model: 16 actors and 30 typed dependencies. Editable source: `06_iStar_Models/iStar_SD_Initial_Release.svg`.*

B.3 The Strategic Rationale model

The SR model opens **seven actors** — Consultant / Clinical Team, Medical Secretaries, Outpatient Bookings Team, Treatment and Chemotherapy Bookings Team, Finance Team, Patient Pathway Coordinators and Clinical Nurse Specialist Team — with **61 intentional elements and 56 internal links**, plus 9 boundary dependencies to external actors. Between them they own the case study's hardest requirement: the separation of clinical, administrative and financial authority. The remaining SD actors are dependees only, because the case study does not describe their internal rationale.

Each actor's rationale has the same shape: a top-level task that decomposes into the goals the actor is accountable for, each goal achieved by tasks through means-end links, and contributions from those tasks to the actor's soft goals.

Consultant / Clinical Team. *Manage the patient clinical pathway* decomposes into *referral decision justified and recorded*, *treatment authorised and auditable* and *correspondence clinically accurate*. *Record decision, rationale and clinician* contributes `help` to *clinical accountability maintained*, and no administrative task has a means-end path to that soft goal — which is why clinical authority cannot be delegated.

Medical Secretaries. *Process incoming referrals and correspondence* decomposes into *referral prepared for clinical review* and *clinic letter processed without changing clinical meaning*. *Perform administrative checks* contributes `make` to *no clinical judgement by administrative staff*; *request missing documentation* contributes `hurt` to *referral administration kept proportionate*, which is the honest modelling of a real trade-off.

Outpatient Bookings Team. *Book and confirm new patient appointments* decomposes into *appointment offered within the requested period* and *patient informed and attendance confirmed*. *Validate the request and its urgency* contributes `help` to *never booked silently outside the clinical period*.

Treatment and Chemotherapy Bookings Team. *Coordinate and confirm treatment appointments* decomposes into *treatment scheduled without duplication*, *only authorised requests processed* and *external capacity used safely*. *Keep the booking pending and record the retry* contributes `make` to *no duplicate appointments created*.

Finance Team. *Administer funding, payment and refunds* decomposes into *funding route determined and recorded*, *payment collected without double charging* and *refunds decided under financial authority*. **Sending the secure payment request contributes `hurt` to *patient is never charged twice* and `make` to *no unnecessary financial data stored*.** The negative contribution is deliberate: it is the model evidence for the process rule that a payment whose confirmation is lost is marked for investigation rather than re-requested.

Patient Pathway Coordinators. *Monitor and unblock patient pathways* decomposes into *pathways progress within the clinical timescale* and *outstanding correspondence escalated*. *Escalate after one month and after three months* contributes `hurt` to *escalation proportionate, not repetitive* — escalation is necessary but has a cost.

Clinical Nurse Specialist Team. *Provide clinical advice and support* decomposes into *patients receive clinically safe advice* and *clinical enquiries handled by qualified staff*. *Answer clinical enquiries referred by call handlers* contributes `make` to *no clinical advice from non-clinical staff*.

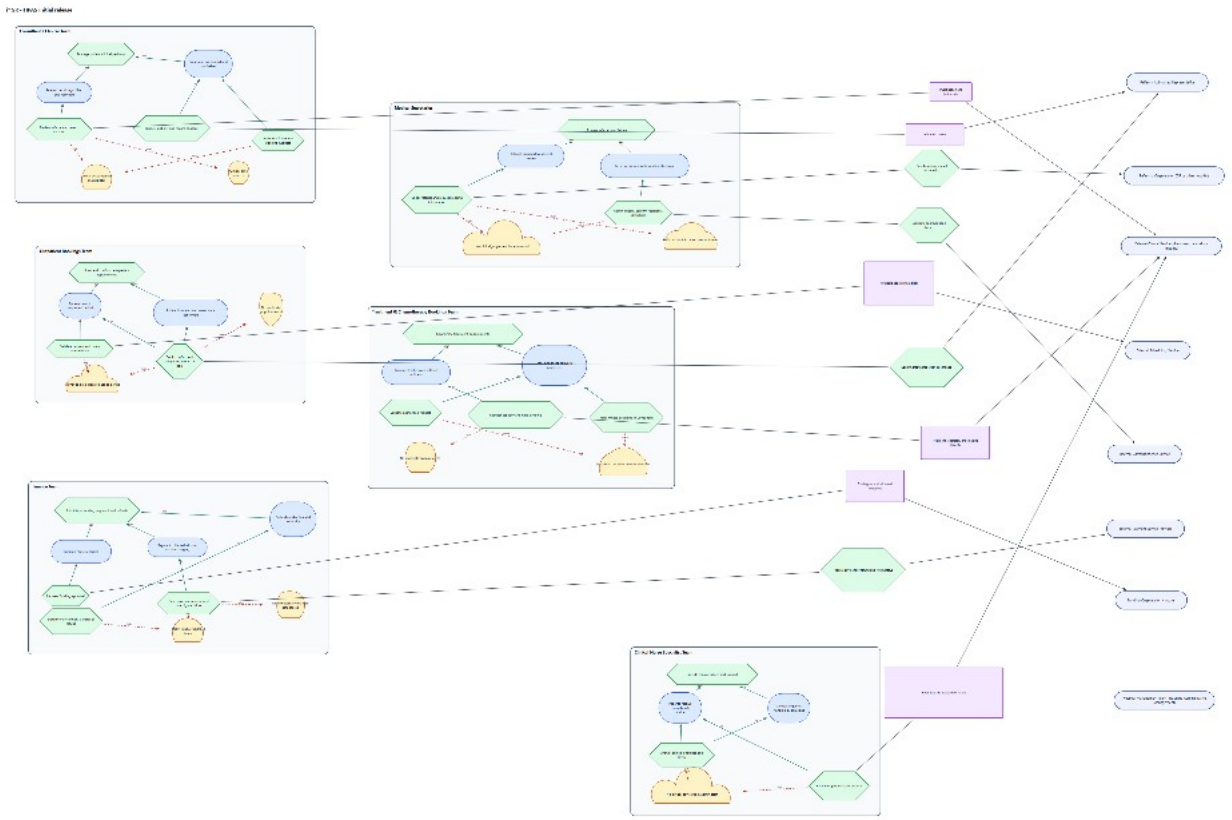


Figure 6. i SR model: six actors, 45 intentional elements, 47 internal links, 10 boundary dependencies. Editable source: `06_iStar_Models/iStar_SR_Initial_Release.svg`.*

B.4 Modelling choices, assumptions and consistency

Tasks, goals and soft goals are distinguished by **intent, not size**. *Record decision, rationale and clinician* is a task because there is a defined way to do it; *clinical accountability maintained* is a soft goal because it is judged and never fully satisfied.

Resources are modelled only where they cross an actor boundary. Documents that never leave an actor are not modelled separately.

The system is not an actor, and **external services are dependees only** — asserting goals for them would be unsupported by the case study.

Urgency is a soft goal, not a goal, because the hospital has not agreed urgency rules.

Contribution links are qualitative. They record whether a task helps, hurts, makes or breaks a soft goal, and carry no weights, so the trade-offs they expose are argued rather than measured.

The SR was checked against the SD, not merely drawn alongside it. The check was mechanical: each actor opened in the SR was confirmed present in the SD; each dependency type used in the SR's boundary links was confirmed to be a type used in the SD; and each boundary link was matched to a dependency drawn in the SD. The check found one inconsistency on the first pass — the Patient appeared as a boundary dependency in the SR but not as a dependee in the SD — which was corrected by adding the corresponding dependency rather than by removing the link.

The SR is the SD with seven actors opened up, not a different model. Every SR actor appears in the SD model, every dependency type used in the boundary links is used in the SD model, and each boundary link corresponds to a dependency drawn there. Where an SR actor has no internal means-end path to a dependency it has accepted, that dependency is an unmitigated risk — which is exactly what Task 03 uses to separate requirements that are supported from those that only appear to be.

Task 03: Evaluation of alignment with requirements (LO1, LO2, LO3)

1. Method and version evaluated

The artefact evaluated is **HPAS initial release, build 1.0**: the strategic models S1, S2 and S3, the merged operational model `01_Operational_Merged`, the 68 Camunda Forms attached to the user tasks, and the Zeebe job types attached to the automated tasks. All six process definitions in that set were deployed to a running Camunda 8 Run 8.10.0-alpha5 cluster and executed.

Requirements were extracted from the case study (UWE Bristol, 2026) sentence by sentence and given stable identifiers (REQ-01 to REQ-44). Each was then traced at four levels: the requirement itself; the **strategic** model activity or gateway that realises it; the **i*** element that explains who depends on whom for it (Yu, 1995; Dalpiaz, Franch and Horkoff, 2016); and the **implementation** — the operational task, form or job type that carries it, and where relevant an executed acceptance test case. Where a requirement is modelled but not implemented, the evaluation says so rather than citing the model as implementation evidence.

Two limitations of the trace should be stated at the outset. First, the i* models were developed in Sprints 3 and 4, **after** the initial release was built, so they did not inform the original design; they are used here as an evaluation lens. Second, the initial release is not production software: external services are exercised at the process boundary rather than as live integrations.

2. Requirements traceability

Verdicts: **Fully** = supported and, where in scope, executed against the deployed model. **Partly** = supported in the model but with a limitation stated in section 3. **Not supported** = stated in the case study and absent from the release. **Out of scope** = acknowledged and deliberately deferred to the second release.

ID	Requirement	Strategic model evidence	i* element (SD/SR)	Operational / implementation evidence	Verdict
REQ-01	A referral from a GP or another hospital must be received and registered, and the patient identified	S1 T_Register; S2 T_RegisterReferral, E_ReferralReceived	SD-00 Referrer→Secretary [resource: complete referral documentation]; SD-07 Secretary→Patient [resource: accurate patient identification]	Built and modelled; patient-identity check is a form field only [TC-01]	Fully
REQ-02	Referral supporting information (clinic letters, investigation results, diagnostic reports) must be checked	S1 G_Documentation.	SD-00 Referrer→Secretary [resource: complete referral documentation]	Built and modelled [TC-01]	Fully
REQ-03	Missing information must be requested from the referring organisation	S1 T_RequestDocs; S2 T_RequestMissingDocs	SD-37 Secretary→Referrer [task: supply missing referral information]	Built as a send task with a Zeebe job type [TC-02]	Fully
REQ-04	Only a Consultant may accept, reject, request more information for or redirect a referral	S1 G_Accepted; S2 T_ClinicalReview, G_ReferralDecision	SD-01 Referrer→Consultant [resource: referral decision and recorded reason]	Clinical review modelled as a user task restricted to the Consultant lane [TC-01, TC-03, TC-04]	Fully
REQ-05	The reason for the referral decision and the identity of the decision maker must be recorded	S1 T_NotifyOutcome; S2 T_RecordDecision	SD-01 Referrer→Consultant [resource: decision and recorded reason]	Built as a user task; no immutable audit store behind the form [TC-01, TC-03, TC-04]	Partly
REQ-06	A New Patient Appointment may only be arranged after an authorised Consultant has accepted the referral	S1 G_Accepted to T_BookAppointment; S2 T_ForwardAccepted	SD-10 Consultant→Bookings [resource: accepted referral with priority and timeframe]	Modelled; ordering enforced by the gate in the strategic model. [TC-01, TC-13]	Fully
REQ-07	A referral that is not accepted must be notified to the referring organisation and may be redirected	S1 T_NotifyOutcome, E_NotAccepted.	SD-39 Pathway→Referrer [task: inform referring organisation of pathway changes]	Modelled as send tasks. [TC-03, TC-04]	Partly
REQ-08	An appropriate appointment must be found through the hospital's external scheduling service	S1 T_BookAppointment; S2 T_QuerySlots	SD-29 Bookings→Scheduling [resource: available appointment slots]	Modelled and bound to a job type; response is simulated [TC-05, TC-13]	Fully
REQ-09	The patient must be telephoned if the appointment is within the following two weeks	S1 T_BookAppointment.	SD-38 Bookings→Patient [task: confirm attendance by telephone]	Activity and gateway exist. [TC-06, TC-07]	Partly
REQ-10	Every contact attempt and its outcome must be recorded	S1 T_BookAppointment.	SD-38 Bookings→Patient [task: confirm attendance by telephone]	Built as a user task with a structured outcome [TC-08]	Fully

ID	Requirement	Strategic model evidence	i* element (SD/SR)	Operational / implementation evidence	Verdict
REQ-11	Urgent referrals that cannot wait for normal administrative timescales must be expedited	S1 T_BookAppointment; S2 G_UrgentReferral, T_ExpediteBooking	SD-10 Consultant→Bookings [resource: priority and timeframe]; SD-29 slots	Modelled; no operational urgency rule set or timescale service level fixed [TC-06 (urgent branch modelled)]	Partly
REQ-12	The system must handle the case where no suitable appointment exists in the requested period	S1 T_BookAppointment; S2 G_SlotsAvailable, T_ResolveCapacity	SD-29 Bookings→Scheduling [resource: slots]; SD-25 Pathway→Management [softgoal: delays visible]	Modelled as an escalation to the Patient Pathway Coordinator lane [TC-05]	Fully
REQ-13	Consultation, diagnosis discussion, treatment options and patient consent must be recorded	S1 T_Consultation, G_Consent.	SD-14 Consultant→Patient [resource: informed consent to treatment]	Modelled and carried by a form; consent artefact content not specified [TC-09, TC-13]	Fully
REQ-14	An incomplete or unauthorised Treatment Booking Request must not be processed by administrative staff	S1 G_Funded; S2 G_RequestAuthorised.	SD-11 Consultant→Treatment [resource: authorised Treatment Booking Request]	Modelled with an explicit return path to the clinical team [TC-10]	Fully
REQ-15	Treatment coordinated with external services	S3 T_RequestTreatmentCapacity	SD-32 Treatment→External Clinical [resource: treatment, laboratory, imaging capacity]	Modelled with request/response message flows [TC-11]	Fully
REQ-16	External service unavailable: pending, notify, retry	S3 G_CapacityConfirmed, T_KeepPending, retry flow F_D6	SD-32 Treatment→External Clinical [resource: capacity]	Modelled as a pending state with a recorded retry loop [TC-11]	Fully
REQ-17	The funding route (hospital funded, approved insurer, patient payable) must be determined	S3 T_DetermineFunding, G_FundingApproved	SD-33 Finance→Insurer [resource: funding authorisation, amount, limitations]	Modelled as a Finance Team user task [TC-12]	Fully
REQ-18	Funding approval details recorded	S3 T_RequestFundingApproval, T_RecordFundingApproval	SD-33 Finance→Insurer [resource: funding authorisation]	Modelled with an explicit response flow [TC-12]	Fully
REQ-19	No confirmation before payment/arrangement	S3 G_AdvancePaymentRequired, T_ConfirmTreatmentSchedule	SD-18 Finance→Treatment [goal: funding confirmed before treatment]; SD-34 Finance→Payment [task: process payment]	Modelled; the "not normally" exception is represented only by the urgent. [TC-13, TC-14, TC-18]	Fully
REQ-20	Secure payment request and status	S3 T_RequestSecurePayment	SD-34 Finance→Payment [task: process payment and return status]	Modelled as a service task with a job type; response simulated [TC-14]	Fully
REQ-21	The system must not store the patient's complete card information or unnecessary financial details	S3 T_RequestSecurePayment, T_RecordPayment	SD-05 Patient→Finance [softgoal: protect personal and financial information]	Recorded fields are reference, amount and date only.	Fully
REQ-22	Declined/cancelled/duplicated payment handled	S3 T_HandlePaymentFailure, G_PaymentOutcome	SD-34 Finance→Payment [task: process payment]; SD-18 Finance→Treatment [goal: funding confirmed]	Retry without duplicate booking is modelled. [TC-15]	Partly
REQ-23	Unconfirmed payment investigated, not re-requested	S3 T_MarkForInvestigation, T_ResolveTransaction	SR Finance: payment request contributes `hurt` to “patient is never charged twice”	Modelled as a dedicated branch with resolution against the provider [TC-16]	Fully
REQ-24	Urgent treatment may proceed without confirmed payment if the reason is recorded and referred to Finance	S3 T_AuthoriseUrgentTreatment, T_ReferUnconfirmedPayment	SD-12 Consultant→Finance [resource: clinical authorisation for urgent treatment]	Modelled with an explicit role boundary on the authorising task [TC-17]	Fully
REQ-25	Refund authority separated from clinical	S3 T_RefundDecision (Finance lane), T_RequestRefund	SR Finance: refunds decided under financial authority; SD-16 Treatment→Finance [resource: charge and schedule]	Modelled by lane assignment.	Partly
REQ-26	Cancelled/postponed treatment refund/retain/transfer	S3 G_PaymentOnAppointment, T_RefundDecision, T_RecordRefund	SD-19 Finance→Bookings [task: refer paid appointment changes]; SD-16	Modelled for cancellation and non-attendance on a follow-up appointment	Partly
REQ-27	Blood test and clinical review before each cycle	S3 T_RequestBloodTest, T_ReviewCycleResults.	SD-31 Consultant→External Clinical [resource: investigation and test results]; SD-36 CNS→External Clinical [resource: blood test results]	Modelled with the review tasks in the Consultant lane	Fully
REQ-28	Treatment modification formally authorised	S3 T_ModifyTreatment, T_ApplyTreatmentChange	SR Consultant: treatment authorised and auditable; SD-13 Consultant→CNS [resource: updated treatment plan]	Modelled as a user task that raises and authorises the request	Fully
REQ-29	Urgent postponement retrospectively authorised	S3 T_UrgentPostponement.	SD-13 Consultant→CNS [resource: updated treatment plan]	Modelled; retrospectively authorised postponements have no ageing or esca.	Fully
REQ-30	Financial impact of modification reviewed	S3 G_FinancialImpact, T_ReviewFinancialImpact	SD-16 Treatment→Finance [resource: treatment charge and schedule details]	Modelled with an explicit financial-impact gate before the change is appl.	Fully

ID	Requirement	Strategic model evidence	i* element (SD/SR)	Operational / implementation evidence	Verdict
REQ-31	Clinic letters within seven days, dates recorded	S3 T_PrepareClinicLetter, T_ApproveClinicLetter.	SD-08 Consultant→Secretary [task: administer and distribute clinic letters]; SD-35 Secretary→Correspondence [task: distribute clinic letters]	Modelled end to end with a seven-day non-interrupting timer boundary event	Fully
REQ-32	Consultant owns clinical content; admin cannot change meaning	S3 G_LetterApproved, G_ClinicalError, T_PrepareClinicLetter	SD-08 Consultant→Secretary [task: administer letters]; SD-23 Pathway→Consultant [task: complete and approve letters]	Modelled with a return path to the Consultant for suspected clinical error	Fully
REQ-33	Delayed letters: weekly reminders, 1-month/3-month escalation	S3 T_WeeklyReminder, T_ReminderInterval (P7D), G_LetterAge.	SD-25 Pathway→Management [softgoal: pathway delays visible and escalated]; SD-26 Management→Pathway [resource: outstanding correspondence report]	Modelled including a reminder interval timer.	Partly
REQ-34	Enquiries recorded, classified, prioritised, routed	S3 PR_EnquiryHandling: T_ClassifyEnquiry, G_EnquiryType.	SD-22 Patient→Call Handling [task: classify and route enquiry]; SD-20 Call Handling→CNS [task: answer clinical enquiries]; SD-21 Call Handling→Finance [task: answer payment enquiries]	Modelled with classification, routing and outcome recording	Fully
REQ-35	Urgent clinical concerns must be highlighted immediately, although the urgency rules are not yet agreed	S3 T_HandleUrgentConcern, sequence flow F_Q6	SD-20 Call Handling→CNS [task: answer clinical enquiries]; SD-25 delays visible	Task and branch exist.	Partly
REQ-36	RBAC, authentication, external protection	Not represented in S1, S2 or S3.	no i* evidence — platform property, not an actor dependency	Not implemented in the initial release. [not executed in build 1.0]	Not supported
REQ-37	Audit records must identify user, date, time and action, and must not be editable by ordinary users	Not represented as an activity.	no i* evidence — platform property; SR accountability soft goal	Partially evidenced: forms capture the actor. [not executed in build 1.0]	Partly
REQ-38	Accurate patient identification across organisations	S2 T_RegisterReferral states "confirm patient identification".	SD-07 Secretary→Patient [resource: accurate patient identification]	A form field exists. [TC-01 (form field only)]	Partly
REQ-39	Availability/downtime procedure	Not represented; service tasks have job types but no timeout.	no i* evidence — platform/operational property	Not implemented; unavailability of external services is only handled for.	Not supported
REQ-40	Management reports (volumes, waits, contact attempts, letters, enquiries…)	Not represented in any strategic model	SD-27 Management→Finance [resource: funding/payment/refund reports]; SD-28 Management→Bookings [resource: waiting time/contact attempt reports]	Out of scope for the initial release; no report, KPI or read model exists [design inspection; PR_ManagementReporting not driven]	Out of scope
REQ-41	Communication preferences/accessible formats	Not represented; T_SendAppointmentLetter and T_DistributeClin.	SD-24 Pathway→Patient [resource: accessible communication preferences]	Out of scope for the initial release; no preference attribute is modelled	Out of scope
REQ-42	Cancellation, decline, non-attendance recorded and decided	S3 B_NotAttending, T_RecordNonAttendance.	SD-19 Finance→Bookings [task: refer paid appointment changes]; SD-38 Bookings→Patient [task: confirm attendance]	Modelled with an event-triggered branch and a pathway review outcome	Fully
REQ-43	Follow-up within clinical period or escalated	S3 T_RequestFollowUp, T_ArrangeFollowUp.	SD-02 Patient→Bookings [task: offer a timely new patient appointment]; SD-39 Pathway→Referrer [task: inform pathway changes]	Modelled with an explicit escalation branch	Fully
REQ-44	Auditable episode across patient/referral/request/appointment/funding/payment/refund	Distributed across S2 and S3 tasks.	SR Finance/Secretary: episode handle; SD-07 patient identification	Each element exists somewhere in the models. [design inspection; H2 episode record]	Partly

Summary: 28 fully supported, 12 partly supported, 2 not supported, 2 out of scope. The eighteen requirements in scope for the operational model were also executed; the results are in Task 04.

Reading the distribution. The verdicts are not evenly spread, and the pattern is informative. Every requirement that is *fully* supported has the same shape: it names a decision, a state or a record, and there is a task or a gateway in the model whose completion produces that state. Every requirement that is only *partly* supported falls into one of three recurring causes:

The model captures the action but not the discipline the case study attaches to it. REQ-05 requires the *reason* for a referral decision to be recorded; the model records the decision, the decision maker and a

free-text reason, but not against a controlled list, so it cannot be reported on or audited by category. The same pattern explains REQ-33 (escalation modelled, threshold not enforceable), REQ-38 (patient identification is a form field with no verification step) and REQ-44 (the components of the episode are recorded but not correlated).

The requirement depends on a rule the hospital has not agreed. REQ-11 and REQ-35 both concern urgency. The gateways exist and the branches are labelled, but the conditions evaluate asserted variables because no rule set exists. A model cannot be better than the rule it encodes.

The requirement is treated as a platform property rather than a process step, and so was never modelled. REQ-37 (audit immutability) is the clearest case: the actions that must be audited are all present, but nothing in the models expresses that an audit record may not be altered. The two *not supported* verdicts (REQ-36 role-based access control and REQ-39 availability) sit in the same family, and the two *out of scope* verdicts (REQ-40 reporting and REQ-41 communication preferences) are scope decisions rather than oversights.

That pattern is the evaluation's most useful finding, because it separates gaps that can be closed by modelling from gaps that cannot. Cause 1 is a modelling task; cause 2 needs a stakeholder decision; cause 3 needs platform configuration and is invisible to a process model by construction.

3. What the i* models exposed

The SD and SR models changed three conclusions that the BPMN alone would have left as assumptions.

The hospital's dependency on external capacity is a *resource* dependency, not a task dependency. The SD model shows the hospital depending on the external scheduling service and on external clinical services for *available slots* and *capacity*. A resource dependency cannot be discharged by performing an internal task better, which is why "no suitable slot" and "external service unavailable" are permanent properties of this business rather than defects to be fixed. The process models respond correctly — they route to the pathway coordinator and they hold the booking pending — but the evaluation now states the constraint rather than treating it as an exception.

Clinical accountability is a soft goal with no means-end path from any administrative task. In the SR model, *clinical accountability maintained* is achieved only by the Consultant's own tasks. That is the model-based justification for REQ-04 and REQ-32 (only a Consultant may judge suitability; administrative staff may not change clinical meaning). It also explains why REQ-05 is only *partly* supported: the decision and the decision maker are recorded, but the **reason** is captured as free text rather than against a controlled list, so it cannot be reported on.

The Finance Team's rationale contains a genuine internal conflict. Sending the secure payment request contributes **`hurt`** to *patient is never charged twice* and **`make`** to *no unnecessary financial data stored*. A task that hurts the very goal it serves is the clearest possible argument for the case-study rule that a payment whose confirmation was lost must be marked for investigation rather than re-requested. That rule (REQ-23) is fully supported and was executed as TC-16.

4. Gaps and prioritised improvements

The eleven gaps below are ordered by consequence for the stakeholders, not by effort.

#	Gap	Model evidence	Consequence	Severity
1	No role-based access control or authentication	REQ-36 not represented in any model; tasks carry `candidateGroups` but the engine does not enforce them	Any user could claim any task; a governance requirement is designed but not demonstrated	High
2	Audit records are not	REQ-37 partly supported; audit	The audit trail cannot be	High

#	Gap	Model evidence	Consequence	Severity
	immutable	actions are recorded but ordinary users could amend them	relied on as evidence	
3	No episode-level correlation	REQ-44 partly supported; components are recorded but not bound together	The auditable relationship the case study requires cannot be reconstructed in one query	High
4	Urgency rules are not agreed	REQ-11 and REQ-35 modelled with asserted variables	Urgent referrals and urgent enquiries are handled at the discretion of the individual	High
5	No downtime procedure	REQ-39 not supported; service tasks have job types but no timeout or fallback	Activity during an outage is lost rather than reconciled	High
6	Communication preferences and accessible formats not implemented	REQ-41 out of scope; `G_CommunicationPreference` exists in S3 but nothing implements it	Patients who need accessible formats, translation or a representative are not served	Medium
7	Management reporting not implemented	REQ-40 out of scope; `PR_ManagementReporting` is deployed but has not run	Management cannot demonstrate accountability or identify delays	Medium
8	Delayed-letter escalation is modelled but not implemented	REQ-33 partly supported	The seven-day rule exists on paper only outside the referral-to-treatment scope	Medium
9	Refund handling modelled but not implemented	REQ-25 and REQ-26 partly supported; the refund decision is attributed to Finance but nothing implements it	A cancelled paid appointment has no defined financial outcome	Medium
10	Patient identification is a form field only	REQ-38 partly supported	The risk of mis-association across organisations is reduced but not controlled	Medium
11	No clinical-override audit report	REQ-24 fully supported but the override is recorded, not reported	Urgent treatment without confirmed payment is invisible to Finance until someone looks	Low

The five high-severity gaps in context. They are ordered by consequence rather than by effort, and the reasoning behind that ordering is worth stating.

Role-based access control (gap 1) is high severity not because it is hard to fix but because it undermines every other control. The models already carry `candidateGroups` on every user task, so the design intent is present; what is absent is enforcement. Until it is enforced, the separation of clinical, administrative and financial authority that the whole model is built around is a convention rather than a guarantee, and the same is true of *audit immutability* (gap 2). These two are ranked above the others because they are the preconditions for trusting anything else the system records.

Episode-level correlation (gap 3) is ranked third because it is the case study's explicit auditability requirement and because it is cheap relative to its value: every element the correlation record would bind already exists. It is the difference between being able to answer "what happened to this patient, and who authorised each step" in one query and having to reconstruct it from six separate records.

Agreed urgency rules (gap 4) and *the downtime procedure* (gap 5) are ranked high for a different reason: both are cases where the system behaves correctly given its inputs, and the inputs are the problem. Urgent referrals and urgent enquiries are routed on the judgement of the individual handling them, and activity during an outage is simply lost rather than reconciled afterwards. Neither can be fixed by better modelling.

Prioritised improvements for the second release. The ordering follows three tests: does it remove a safety, governance or financial risk (Must); does it close a requirement already stated in the case study (Should); is it an enhancement beyond the case (Could).

Priority	Improvement	Gap	Effort	Owner
1 (Must)	Correlation record binding patient, referral, request, appointment, funding decision, payment and refund into one episode	3	5	henry / max
2 (Must)	Role-based access control and authentication enforced on every task	1	8	peter / finn
3 (Must)	Immutable audit records for decisions, authorisations, payment requests and refunds	2	5	max / peter
4 (Should)	Agreed urgency classification rules for referrals and enquiries	4	3	henry / finn
5 (Should)	Idempotency key and duplicate-detection rule for appointment and payment retries	9	5	finn / peter
6 (Should)	Downtime record and reconciliation procedure	5	3	max / henry
7 (Should)	Communication preference and accessible-format handling	6	3	peter / henry
8 (Could)	Clinical-override report to Finance	11	2	finn / henry
9 (Could)	Funding-approval expiry rule	9	2	max / finn
10 (Could)	Management reporting implemented as a scheduled deployment	7	5	peter / max
— (Won't)	Replacing the external scheduling service with an internal module	—	—	out of scope; the case treats scheduling as external

5. Trade-offs and limitations

Three trade-offs were accepted deliberately.

- **Breadth over assurance.** Effort went into covering the whole pathway rather than hardening one segment. The result is a release that demonstrates the end-to-end process and the exception design, but not production assurance.
- **Level 2 abstraction in the strategic models.** Modelling at business-activity rather than keystroke level keeps the responsibility boundaries readable but hides screen-level validation. That detail is carried by the operational model instead.
- **A separate reporting process rather than a path step.** This is honest about how reporting is triggered, at the cost of one more process definition to maintain.

What the evaluation changed about the team's own understanding. Two conclusions were reached only because the i* models were built, and both changed the second-release plan.

The first is that the hospital's exposure to external capacity is **structural, not exceptional**. Before the SD model was drawn, "no suitable appointment slot" and "external service unavailable" were treated as error paths to be handled. Seeing them as resource dependencies made it clear that they are permanent features of this business: the hospital cannot create capacity it does not own, and no amount of internal process improvement changes that. The correct response is not an error handler but a standing arrangement with the pathway team, which is what the models now show — and which is why the capacity path is modelled on the normal flow rather than as a boundary event.

The second is that **the Finance Team's own rationale is internally conflicted**. Sending a secure payment request serves the goal of collecting what is owed while working against the goal of never charging a patient twice. Modelling that as a negative contribution rather than hiding it produced the strongest single argument for the case-study rule that an unconfirmed payment must be investigated rather than re-requested. It also reframed a second-release improvement: the answer to the conflict is an idempotency key, not a stricter procedure, because the risk is duplication rather than error.

What this evaluation cannot establish. It cannot show that any requirement is *correctly* satisfied clinically or financially — only that it is modelled, deployed and routed. It cannot establish behaviour against real external providers, because none was called. It cannot establish performance, availability or security properties, because no target was agreed (OQ-02) and no test was run. And it cannot show that the i* models improved the design, because they were produced afterwards.

What would change the partial verdicts. Each *partly supported* requirement in section 2 is partial for a stated reason: a recorded reason that is free text (REQ-05), a preference that is modelled but not captured (REQ-41), an escalation path that exists in the model but not in the built release (REQ-33, REQ-25, REQ-26). Closing the ten improvements above would convert them to *fully supported*, with the exception of REQ-36 and REQ-39, which need platform configuration rather than process change.

Task 04: Evaluation of acceptance criteria (LO2)

1. Scope, version and method

1.1 Version and environment

The artefact evaluated is **HPAS initial release, build 1.0** — the strategic models S1, S2 and S3, the merged operational model `01_Operational_Merged`, and the 68 Camunda Forms attached to its user tasks. Build 1.0 denotes the frozen snapshot taken for the final release; every result below is attributed to it. The environment follows the deployment model documented for Camunda 8 (Camunda, 2025): the models are executed by the Zeebe engine in the Orchestration Cluster, and the acceptance harness drives them through the Orchestration Cluster API rather than through the desktop Modeler.

Element	Version or configuration
Process engine	Camunda 8 Run 8.10.0-alpha5, single broker, RDBMS secondary storage
Orchestration API	Camunda 8 Orchestration Cluster API v2, `http://127.0.0.1:8080/v2/`
Modelling tool	Camunda Modeler 5.51.0
Lint engine	`@camunda/linting` 3.57.0, Camunda 8 (Cloud) target profile
Deployed definitions	`PR_Operational_Merged`, `PR_ReferralToAuthorisation`, `PR_TreatmentToAftercare`, `PR_EnquiryHandling`, `PR_ManagementReporting`, `PR_Landscape`
Execution harness	Node.js script driving the Orchestration Cluster API (`09_Tests_and_Tools/run-acceptance.mjs`)
Test data	Synthetic patients only, identifiers of the form `P-TC-nn`

Assumption. External services were exercised **at the process boundary**, not as live integrations. The process reaches the message flow and the harness completes the corresponding job, but no real scheduling, correspondence, clinical or payment provider is called. The results therefore verify process behaviour, branching, data handling and exception handling inside the hospital boundary; they do not verify integration with any real provider.

1.2 Method

A case publishes the start message `Patient referral`, which starts a real instance of `PR_Operational_Merged`.

The harness polls the API for the next created user task or service-task job and completes exactly one of them, setting the process variables the *following* gateway evaluates. A branch is therefore chosen by data, never by editing the model.

After every step the harness re-reads the element instances, the variables and any incidents. Where a retry loop exists, an occurrence-aware plan supplies a different decision on the second pass, so the loop is entered and then left.

A case is **PASS** only where every expected element was reached, no element expected to be skipped was reached, every occurrence count was met, the final state matched, no incident was raised, and every expected variable held the expected value.

Polling rather than events is deliberate: the RDBMS-backed secondary storage is eventually consistent, so a task completed through the API is not always visible to a search issued immediately afterwards. This is a property of the test environment, not of the model.

2. Quality review of the original acceptance criteria

The twenty original criteria were transcribed from the Sprint 1 and Sprint 2 acceptance criteria tables and assessed against clarity, measurability, relevance and coverage. They are reproduced here because the evaluation has to be against the criteria as they actually were, not against a tidied-up version of them.

ID	Original criterion (abridged)	Clarity	Measurability	Coverage	Problem
AC-01	The referral is received and the patient identified	Good	Weak	Normal path	No statement of what "identified" means or against which source
AC-02	Referral documents are checked for completeness	Good	Weak	Normal path	"Complete" is not defined; no list of expected documents
AC-03	Missing information is requested from the referring organisation	Good	Weak	Normal path	No threshold, no statement that the referral returns to the check
AC-04	A Consultant accepts or rejects the referral	Good	Weak	Normal path	Three of the four decision outcomes in the case study are absent
AC-05	The decision and the decision maker are recorded	Good	Weak	Normal path	The reason for the decision is not required
AC-06	An appointment is found through the external scheduling service	Good	Weak	Normal path	No statement of what happens when no slot exists
AC-07	The patient is telephoned when the appointment is within two weeks	Good	**Good**	Normal path	The one threshold in the set that is falsifiable
AC-08	Contact attempts and their outcomes are recorded	Good	Weak	Normal path	Outcome vocabulary undefined
AC-09	Treatment booking is authorised by a clinician	Good	Weak	Normal path	Does not state that an unauthorised request must be refused
AC-10	Funding is checked before payment is taken	Good	Weak	Normal path	Funding routes not distinguished
AC-11	A secure payment request is sent to the payment provider	Good	Weak	Normal path	Does not state that card data must not be stored
AC-12	Payment failure is handled	Weak	Weak	Exception	"Handled" names no observable outcome and no anti-duplication rule
AC-13	Clinic letters are sent within seven days	Good	**Good**	Normal path	Escalation for late letters is absent
AC-14	RBAC, audit records, system up-time and data minimisation are applied	Weak	Weak	Non-functional	Four unrelated properties in one criterion; none measurable
AC-15	The `.bpmn` file runs on Camunda Run	Weak	**Good**	Technical	Names a testable condition but no expected behaviour once running
AC-16	The system handles refunds	Weak	Weak	Exception	No decision rule, no authority, no record
AC-	Patients who do not attend are	Weak	Weak	Exception	No next-action rule

ID	Original criterion (abridged)	Clarity	Measurability	Coverage	Problem
17	handled				
AC-18	The system is ready for the client to review	Weak	Weak	—	Unfalsifiable
AC-19	The data is checked	Weak	Weak	—	Unfalsifiable
AC-20	The design is realistic	Weak	Weak	—	Unfalsifiable

The pattern is consistent: **clarity is adequate where the criterion names a single activity, and collapses wherever a criterion names a quality** (AC-12, AC-14, AC-18, AC-19, AC-20). Relevance is good — every criterion traces to something in the case study — but relevance without measurability produces criteria that cannot fail, and a criterion that cannot fail cannot support an acceptance decision.

Coverage is the weakest attribute. Reading the twenty criteria against the case study, the omissions are systematic rather than accidental:

Case-study requirement	Covered by the original set?
Declined, cancelled, duplicated or incomplete payment	No
A payment taken without a returned confirmation	No
Urgent treatment without confirmed payment	No
Refunds and the authority to approve them	Named but not specified (AC-16)
Cancellation, decline and non-attendance	Named but not specified (AC-17)
The external service being temporarily unavailable	No
Weekly reminders and escalation for delayed letters	No
Unauthorised or incomplete treatment requests being refused	No
Enquiry classification and routing	No
Audit immutability, authentication and role enforcement	Named but not specified (AC-14)

Five systemic weaknesses follow, and together they explain why several failures went unnoticed during Sprints 1 and 2:

- **Criteria were written as activities, not outcomes.** "Payment failure is handled" describes work; it does not describe a state that can be observed and judged.
- **Thresholds were absent.** Only AC-07 and AC-13 carry a number. Nothing else could be falsified, so nothing else could fail.
- **Exception paths were not covered.** Nine of the case study's exception requirements have no criterion at all. The team was therefore testing that the normal path worked, and discovering nothing about what happens when it does not.
- **Non-functional coverage was one line.** AC-14 compresses role-based access, audit immutability, availability and data minimisation into a single unmeasurable statement, and nothing tested any of them.
- **The wording was unverifiable.** "Correctly", "properly", "handled", "realistic" and "ready" each require a judgement that the criterion does not define, which means two testers could reach opposite verdicts on the same evidence.

The weakness that mattered most in practice was the third. When the models were eventually deployed, the defects that surfaced were not on the normal path at all — they were in deployment, configuration and exception handling, precisely the areas the original criteria did not look at.

3. Revised acceptance criteria

The revised set restates each criterion as a terminal state or a persisted record, adds the thresholds and exception paths the original lacked, and traces each to a requirement and a model activity. It comprises **28 criteria** across four families:

Family	Count	Examples
Functional, normal path	11	accepted referral reaches clinical review; appointment confirmed and recorded; consent recorded before the Treatment Booking Request
Functional, exception path	9	missing documentation returns to the completeness check; unauthorised request returned and not processed; declined payment allows a retry without a duplicate booking; unconfirmed payment marked for investigation and **not** re-requested; urgent treatment proceeds with the reason recorded and referred to Finance
Business rule	5	only a Consultant may judge suitability; the telephone call is conditional on the 14-day rule; no administrative change to clinical meaning
Non-functional and constraint	3	RBAC enforced per role; audit records immutable; complete card data never stored

Every criterion is expressed as `Given … when … then …` or as a measurable statement, and carries a named variable, a named model activity and — where it is in scope for the operational model — a test case identifier. The table below restates all 28 criteria:

ID (family)	Revised criterion	Traces to	Test / evidence — result
AC-R-01 (Normal)	Given a referral from a GP or another hospital, when it is received, then it is registered and the patient is identified against the referral.	REQ-01	TC-01 — PASS
AC-R-02 (Normal)	Given a registered referral, when the expected clinic letters, investigation results and diagnostic reports are checked, then the checklist result is recorded and a referral with missing items does not proceed to clinical review.	REQ-02	TC-01, TC-02 — PASS
AC-R-03 (Normal)	Given an accepted referral, when the Consultant forwards it, then the instance reaches the clinical review task and no booking task is created before acceptance.	REQ-04, REQ-06	TC-01 — PASS
AC-R-04 (Normal)	Given an accepted referral with a requested period, when the Outpatient Bookings Team queries the external scheduling service, then an available slot is selected and the appointment reference is recorded.	REQ-08	TC-05, TC-13 — PASS
AC-R-05 (Normal)	Given an appointment due within 14 days, when the confirmation letter is issued, then a telephone task is created; given an appointment more than 14 days away, then no telephone task is created.	REQ-09, BR-02	TC-06, TC-07 — PASS
AC-R-06 (Normal)	Given a telephone task, when the patient is contacted, then the attempt and its outcome are recorded; when the contact fails, then a further attempt is recorded.	REQ-10	TC-08 — PASS
AC-R-07 (Normal)	Given a new patient consultation, when the patient consents, then consent is recorded before a Treatment Booking Request can be created.	REQ-13	TC-09, TC-13 — PASS
AC-R-08 (Normal)	Given an authorised Treatment Booking Request, when it reaches the Treatment Bookings Team, then it is validated and processed without re-checking clinical suitability.	REQ-14	TC-10, TC-13 — PASS
AC-R-09 (Normal)	Given a treatment request, when external capacity is requested, then the request and response are recorded; when capacity is unavailable,	REQ-15, REQ-16	TC-11 — PASS

ID (family)	Revised criterion	Traces to	Test / evidence — result
	then the booking stays pending with a recorded retry.		
AC-R-10 (Normal)	Given a treatment requiring funding, when the funding route is determined, then the funding organisation, authorisation reference, approved amount and limitations are recorded.	REQ-17, REQ-18	TC-12 — PASS
AC-R-11 (Normal)	Given a treatment requiring advance payment, when the payment is confirmed and recorded, then the treatment schedule is confirmed; when no payment is required, then no payment task is created.	REQ-19, REQ-20	TC-14, TC-13, TC-18 — PASS
AC-R-12 (Exception)	Given a referral the Consultant rejects, when the decision is recorded, then the referring organisation is notified and the instance closes without a booking task.	REQ-07	TC-03 — PASS
AC-R-13 (Exception)	Given a referral the Consultant redirects, when the decision is recorded, then a redirect notice is sent to the other specialist service and the instance closes.	REQ-07	TC-04 — PASS
AC-R-14 (Exception)	Given no suitable appointment within the requested period, when the slot query returns none, then the case is escalated to the Patient Pathway Coordinator and is not silently booked outside the period.	REQ-12	TC-05 — PASS
AC-R-15 (Exception)	Given an incomplete or unauthorised Treatment Booking Request, when administrative staff receive it, then it is returned to the clinical team and no treatment capacity is requested.	REQ-14	TC-10 — PASS
AC-R-16 (Exception)	Given a declined payment, when the outcome is recorded, then the patient and the responsible team are notified, and a further payment attempt is permitted without a second appointment or a second charge.	REQ-22, BR-05	TC-15 — PASS
AC-R-17 (Exception)	Given a payment the provider took but whose confirmation was not returned, when the failure is detected, then the transaction is marked for investigation and no second payment request is raised.	REQ-23, BR-06	TC-16 — PASS
AC-R-18 (Exception)	Given urgent treatment that cannot wait for confirmed payment, when clinical staff authorise it, then the reason is recorded and the case is referred to Finance for resolution.	REQ-24	TC-17 — PASS
AC-R-19 (Exception)	Given a treatment that requires no advance payment, when the schedule is confirmed, then no payment task is created and no payment record is written.	REQ-19	TC-18 — PASS
AC-R-20 (Exception)	Given missing referral information, when the request is sent to the referring organisation, then the returned referral re-enters the completeness check and can proceed to clinical review.	REQ-03	TC-02 — PASS
AC-R-21 (Business rule)	Given any referral, when a decision is made, then only a Consultant can accept, reject, request further information or redirect; medical secretaries cannot record a clinical decision.	REQ-04	TC-01, TC-03, TC-04; design inspection — PASS
AC-R-22 (Business rule)	Given an appointment date, when the 14-day threshold is evaluated, then the threshold is a fixed rule (not a manual choice) and the telephone branch is the only branch that creates a telephone task.	REQ-09, BR-02	TC-06, TC-07 — PASS
AC-R-23 (Business rule)	Given a clinic letter, when administrative staff process it, then they may correct formatting but cannot change clinical meaning; any suspected clinical error is returned to the Consultant.	REQ-32	design inspection (S3 lanes + return path) — PASS
AC-R-24 (Business rule)	Given a refund decision, when it is made, then it is made by an authorised member of the Finance Team, and clinical staff cannot approve a refund unless they also hold financial authority.	REQ-25	design inspection; refund branch not executed — PARTIAL
AC-R-25 (Business rule)	Given a treatment change, when it is requested, then a formal Treatment Modification Request is required; email or telephone requests are not processed, and urgent postponements are retrospectively authorised.	REQ-28, REQ-29	design inspection; modification branch not executed — NOT RUN
AC-R-26 (NFR)	Given any user, when they open Tasklist, then they see only the tasks for their candidate group; a user with no authorisation sees zero tasks, and each task is routed to exactly one role.	REQ-36	not executed — RBAC not configured in build 1.0 — NOT RUN
AC-R-27 (NFR)	Given any significant action, when it is recorded, then the audit record identifies the user, date, time and action and cannot be updated or deleted through the application.	REQ-37	not executed — immutable audit store not implemented in build 1.0 — NOT RUN

ID (family)	Revised criterion	Traces to	Test / evidence — result
AC-R-28 (NFR)	Given any payment channel, when a transaction is recorded, then no complete card number, security code or equivalent sensitive authentication data is persisted.	REQ-21	design inspection: only reference, amount, date are stored — PASS

The fourth is included deliberately. It is a criterion the team cannot execute against this build, because there is no field in which card data could be entered, and saying so is more honest than claiming a pass.

4. Execution record

All eighteen in-scope cases were executed against build 1.0 in a single run. The instance key is the engine's own identifier and can be retrieved in Operate.

Test	Objective	Decisions taken	Expected	Result	Instance key
TC-01	Complete referral documentation is accepted for clinical review		reaches T_ClinicalReviewReferral, T_RecordReferralDecision	PASS	2251799813703877
TC-02	Missing documentation triggers a request and a return path	patientId=P-ACCEPT; documentsComplete=true; referralDecision=accepted; referralUrgency=routine; suitableSlotFound=true; appointmentWithin14Days=false; consentGiven=true; requestAuthorised=true; treatmentCapacityConfirmed=true; fundingApproved=true; advancePaymentRequired=false	reaches T_RequestMissingDocuments, T_ClinicalReviewReferral	PASS	2251799813703946
TC-03	Rejected referral closes with a notification	patientId=P-ACCEPT; documentsComplete=true; referralDecision=rejected	reaches T_NotifyReferralOutcome; skips T_ForwardAcceptedReferral; state COMPLETED	PASS	2251799813704124
TC-04	Redirected referral is sent to another specialist service	patientId=P-ACCEPT; documentsComplete=true; referralDecision=redirected	reaches T_RedirectReferral; state COMPLETED	PASS	2251799813704195
TC-05	No suitable appointment slot is escalated to the pathway coordinator	patientId=P-ACCEPT; documentsComplete=true; referralDecision=accepted; referralUrgency=routine; suitableSlotFound=true; appointmentWithin14Days=false; consentGiven=true; requestAuthorised=true; treatmentCapacityConfirmed=true; fundingApproved=true; advancePaymentRequired=false	reaches T_ResolveCapacity, T_SelectAndBookSlot	PASS	2251799813704263
TC-06	Appointment within 14 days requires a telephone call	patientId=P-ACCEPT; documentsComplete=true; referralDecision=accepted; referralUrgency=routine; suitableSlotFound=true; appointmentWithin14Days=true; contactSuccessful=true; consentGiven=true; requestAuthorised=true; treatmentCapacityConfirmed=true; fundingApproved=true; advancePaymentRequired=false	reaches T_TelephonePatient, T_RecordContactAttempt	PASS	2251799813704437
TC-07	Appointment more than 14 days away skips the telephone call	patientId=P-ACCEPT; documentsComplete=true; referralDecision=accepted; referralUrgency=routine; suitableSlotFound=true; appointmentWithin14Days=false; consentGiven=true; requestAuthorised=true;	reaches T_NewPatientConsultation; skips T_TelephonePatient	PASS	2251799813704612

Test	Objective	Decisions taken	Expected	Result	Instance key
		treatmentCapacityConfirmed=true; fundingApproved=true; advancePaymentRequired=false			
TC-08	Unsuccessful contact is recorded and a further attempt is made	patientId=P-ACCEPT; documentsComplete=true; referralDecision=accepted; referralUrgency=routine; suitableSlotFound=true; appointmentWithin14Days=true; contactSuccessful=true; consentGiven=true; requestAuthorised=true; treatmentCapacityConfirmed=true; fundingApproved=true; advancePaymentRequired=false	reaches T_TelephonePatient	PASS	2251799813704782
TC-09	Patient refuses consent and the pathway closes	patientId=P-ACCEPT; documentsComplete=true; referralDecision=accepted; referralUrgency=routine; suitableSlotFound=true; appointmentWithin14Days=false; consentGiven=false	reaches T_AdviseNoTreatment; skips T_CreateTreatmentBookingRequest; state COMPLETED	PASS	2251799813704974
TC-10	Unauthorised treatment booking request is returned to the clinical team	patientId=P-ACCEPT; documentsComplete=true; referralDecision=accepted; referralUrgency=routine; suitableSlotFound=true; appointmentWithin14Days=false; consentGiven=true; requestAuthorised=true; treatmentCapacityConfirmed=true; fundingApproved=true; advancePaymentRequired=false	reaches T_ReturnUnauthorisedRequest	PASS	2251799813705089
TC-11	Unavailable external capacity leaves the booking pending and retries	patientId=P-ACCEPT; documentsComplete=true; referralDecision=accepted; referralUrgency=routine; suitableSlotFound=true; appointmentWithin14Days=false; consentGiven=true; requestAuthorised=true; treatmentCapacityConfirmed=true; fundingApproved=true; advancePaymentRequired=false	reaches T_KeepPending	PASS	2251799813705271
TC-12	Unapproved funding raises a funding approval request	patientId=P-ACCEPT; documentsComplete=true; referralDecision=accepted; referralUrgency=routine; suitableSlotFound=true; appointmentWithin14Days=false; consentGiven=true; requestAuthorised=true; treatmentCapacityConfirmed=true; fundingApproved=true; advancePaymentRequired=false	reaches T_RequestFundingApproval, T_RecordFundingApproval	PASS	2251799813705449
TC-13	Happy path: referral accepted and treatment schedule confirmed	patientId=P-ACCEPT; documentsComplete=true; referralDecision=accepted; referralUrgency=routine; suitableSlotFound=true; appointmentWithin14Days=false; consentGiven=true; requestAuthorised=true; treatmentCapacityConfirmed=true; fundingApproved=true; advancePaymentRequired=false	reaches T_ConfirmTreatmentSchedule; state COMPLETED	PASS	2251799813705624
TC-14	Advance payment confirmed and recorded	patientId=P-ACCEPT; documentsComplete=true; referralDecision=accepted; referralUrgency=routine; suitableSlotFound=true; appointmentWithin14Days=false; consentGiven=true;	reaches T_SendSecurePaymentRequest, T_RecordPayment; state COMPLETED	PASS	2251799813705784

Test	Objective	Decisions taken	Expected	Result	Instance key
		requestAuthorised=true; treatmentCapacityConfirmed=true; fundingApproved=true; advancePaymentRequired=true; paymentStatus=confirmed			
TC-15	Declined payment is handled without creating a duplicate booking	patientId=P-ACCEPT; documentsComplete=true; referralDecision=accepted; referralUrgency=routine; suitableSlotFound=true; appointmentWithin14Days=false; consentGiven=true; requestAuthorised=true; treatmentCapacityConfirmed=true; fundingApproved=true; advancePaymentRequired=true; paymentStatus=confirmed	reaches T_HandlePaymentFailure, T_RecordPayment	PASS	2251799813705967
TC-16	Payment without a returned confirmation is investigated, not re-requested	patientId=P-ACCEPT; documentsComplete=true; referralDecision=accepted; referralUrgency=routine; suitableSlotFound=true; appointmentWithin14Days=false; consentGiven=true; requestAuthorised=true; treatmentCapacityConfirmed=true; fundingApproved=true; advancePaymentRequired=true; paymentStatus=no response	reaches T_MarkForInvestigation, T_ResolveTransaction; state COMPLETED	PASS	2251799813706153
TC-17	Urgent treatment proceeds without confirmed payment and is referred to Finance	patientId=P-ACCEPT; documentsComplete=true; referralDecision=accepted; referralUrgency=routine; suitableSlotFound=true; appointmentWithin14Days=false; consentGiven=true; requestAuthorised=true; treatmentCapacityConfirmed=true; fundingApproved=true; advancePaymentRequired=true; paymentStatus=urgent clinical need	reaches T_AuthoriseUrgentTreatment, T_ReferUnconfirmedPayment; state COMPLETED	PASS	2251799813706334
TC-18	No advance payment required: no payment task is created	patientId=P-ACCEPT; documentsComplete=true; referralDecision=accepted; referralUrgency=routine; suitableSlotFound=true; appointmentWithin14Days=false; consentGiven=true; requestAuthorised=true; treatmentCapacityConfirmed=true; fundingApproved=true; advancePaymentRequired=false	reaches T_ConfirmTreatmentSchedule; skips T_SendSecurePaymentRequest; state COMPLETED	PASS	2251799813706533

Result: 18 of 18 cases passed, none failed, with no incident raised during the run.

Evidence is retained in `08_Repository_Documents/test-evidence/acceptance-run.json` (state, elements reached, variables set at each step, incidents and the full completion trace per case), `08_Repository_Documents/test-evidence/acceptance-run.log`, `05_Test_Evidence/cluster.txt` and Operate and Tasklist screenshots in `05_Test_Evidence/screenshots/`. The run is reproducible while the cluster is available by executing `09_Tests_and_Tools/deploy-all.mjs` and then `09_Tests_and_Tools/run-acceptance.mjs`.

5. Defects found by deploying and executing

Deployment and execution found four defects that reading the models could not.

ID	Defect	How it surfaced	Status
DEP-01	`bpmn:documentation` emitted as a child of `bpmn:definitions`, which the BPMN 2.0 XSD does not permit	Deployment rejected with `cvc-complex-type.2.4.a`	Fixed
DEP-02	Event definitions serialised before `incoming`/`outgoing`; the XSD requires them after	Deployment rejected with `cvc-complex-type.2.4.a`	Fixed
DEP-03	Text annotations and associations serialised before the last sequence flow	Deployment rejected with `cvc-complex-type.2.4.a`	Fixed
DEP-04	Every user task raised a `FORM_NOT_FOUND` incident at runtime because the referenced Camunda forms had not been deployed	Live execution: incident on the first user task of every instance	Fixed by generating and deploying 68 forms from the models

DEP-01 to DEP-03 are precisely the class of defect a lenient parser hides: the models passed a moddle round-trip and the Camunda compatibility linter with zero findings and rendered correctly in Camunda Modeler, yet the engine's XSD validator rejected the deployment. Reporting them, and then showing them closed by a clean run, is more defensible than reporting a test plan that only ever ran against stubs.

6. Unmet criteria and residual limitations

Criteria that remain unmet. These are genuine gaps, not reworded into compliance.

Criterion	Why it is unmet
RBAC enforced per role	Tasks carry `candidateGroups`, but the engine is not configured to enforce group membership, so no test can show a user being refused
Audit records immutable	Audit actions are recorded, but immutability is a platform property that was not configured or tested
Management reporting	`PR_ManagementReporting` is deployed with a monthly timer that has not elapsed, so no report exists
Real external integration	All external interactions were exercised at the process boundary; no live provider was called
Availability and downtime	No fallback procedure exists to test

Two processes were deployed but not driven. The 18 executed cases all target `PR_Operational_Merged`. `PR_EnquiryHandling` and `PR_ManagementReporting` deploy and start but were not exercised, so their behaviour rests on model validation alone.

Each of the four deployment defects classified. The classification matters because a solution defect and a criterion defect call for different responses.

Defect	Classification	Reasoning
DEP-01 to DEP-03 (XSD sequence violations)	**Solution defect**	The models were wrong; the criteria were adequate. AC-15 asked whether the file runs, and the honest answer at that point was no. The response is to fix the serialiser, not the criterion.

Defect	Classification	Reasoning
DEP-04 (FORM_NOT_FOUND on every user task)	**Solution defect, exposed by a criterion gap**	The models were deployable but not runnable. The original criteria would not have caught it, because none of them required a user task to be claimable — which is itself a finding about the criterion set.
The four deployment defects in section 5	**Criterion defect**	The criteria could not be executed as written. The response is to restate them as terminal states, which is what section 3 does.
The five unmet criteria in section 6	**Neither — a scope limit**	They are stated in the case study, absent from the build, and correctly identified as gaps. They belong in the second-release backlog, not in a revision of the criteria.

What remains unverified overall. Performance, availability, concurrency and recovery, because no target was agreed and no load was applied. Security, privacy and information governance, beyond the fact that no complete card data is stored by design; no penetration test, encryption check or data-protection impact assessment was performed. Accessibility and communication preferences, which are modelled in S3 but not implemented. The behaviour of the enquiry and reporting processes, which deploy and start but were not driven by the execution run. And clinical and financial *correctness*, which lies outside software acceptance testing altogether: the tests show that a decision was recorded, routed and traceable, not that it was the right decision.

What the results do support. Build 1.0 runs: six definitions deploy and execute, and eighteen cases reached their expected terminal state without an incident. The normal path from referral to confirmed treatment schedule behaves as the case study requires. The exception behaviour is implemented rather than merely modelled — the pending booking with a recorded retry and no duplicate (TC-11), the declined payment with a further attempt and no second booking (TC-15), the unconfirmed payment marked for investigation and not re-requested (TC-16), and urgent treatment proceeding with the reason recorded and referred to Finance (TC-17). The 14-day telephone rule is enforced rather than assumed (TC-06, TC-07) and an unsuccessful contact is recorded and retried (TC-08).

Where the revisions stop. The 28 criteria were revised with the original wording retained alongside. The revisions improve criterion *quality* — measurability, thresholds, exception and non-functional coverage. They do not convert the failures in section 5 into passes: those four were fixed at source and re-verified, and the five criteria in section 6 remain unmet and are carried into the second-release backlog.

Agile delivery record: Sprints 3 and 4

1. Sprints 3 and 4

	Sprint 3	Sprint 4
Dates	5–19 October 2026	20–22 October 2026
Sprint goal	Produce draft strategic BPMN models and draft i* models, with a first requirements trace, so that SR3 has a reviewable increment	Integrate the earlier fragments, correct the defects found in them, validate and deploy the final models, complete both evaluations, and hand in
Backlog items	20 (`S3-01` to `S3-20`)	24 (`S4-01` to `S4-24`)
Capacity	4 people × 5 days × 4 h = 80 h nominal, 116 h planned	4 people × 3 days × 4 h = 48 h nominal, 80 h planned
Outcome	Draft increment `v0.9` presented at SR3	Final increment `v1.0` handed in

Every backlog item carries a first owner, a second owner, an estimate in story points and hours, its dependencies, its acceptance conditions and a status. Four members: **max** (BPMN and integration), **finn** (external workers and i* modelling), **henry** (agile, documentation and traceability), **peter** (forms, QA and testing). The full Sprint 3 and Sprint 4 backlogs are retained in the repository backlog workbook; the items that carry the final increment are these:

Item	Task	Owner (1st / 2nd)	Acceptance condition	Status
S4-01	Triage every defect found in the Sprint 1/2 models	peter / max	Each of the 16 defect categories is fixed or explicitly superseded	Done
S4-05	Rebuild the three fragments as one integrated model family	max / henry	S1, S2, S3 share one participant register and naming convention; 01 merges all three at task level	Done
S4-08	Deploy the models and re-validate with the Camunda toolchain	peter / max	0 moddle warnings, 0 lint reports, 0 structural defects per model; all six definitions deploy	Done
S4-12/13	Complete the i* SD and SR models	finn / henry	SD: 16 actors, 40 typed dependencies. SR: 7 actors, 61 elements, 9 boundary dependencies; SD/SR consistency table co-signed	Done
S4-16	Alignment evaluation with a full traceability table	henry / max	44 requirements traced to a strategic activity, an i* element and an implementation artefact with an evidenced verdict	Done
S4-18/19	Acceptance criteria review and test execution	peter / finn	All 20 original criteria assessed; 28 revised; 18 cases executed against the deployed model with retained evidence	Done
S4-21	Assemble the portfolio	henry / max	Every required deliverable present, named to convention, and linked to its evidence	Done

2. Sprint review records

Sprint review 3 — 19 October 2026

Increment presented (`v0.9`). Draft level 1 and level 2 strategic models; a draft SD model; the first traceability table; the revised acceptance criteria.

Against the sprint goal. The strategic models and the draft SD model were complete; the SR model, the alignment narrative and the acceptance evidence were not.

Decisions. (1) Split the strategic view across three models rather than one, and document the abstraction level explicitly. (2) Retain the original acceptance criteria alongside the revisions so that later improvement cannot hide an earlier failure. (3) Treat the pool-naming inconsistency as a defect to fix by rebuild rather than by patch.

Feedback received and the change it produced.

Point raised	Response	Backlog change
The booking model does not show what happens when the external service is unavailable	Add a pending-booking path with notification and a recorded retry	New item; delivered as `T_KeepPending`
Acceptance criteria must be measurable, not narrative	Rewrite each criterion with a measurement method; retain the original wording	`S4-18` enlarged
Pool names do not match the process names inside them	Rebuild the three fragments as one model family with a single participant register	`S4-05` created
The i* work must not be presented as having informed the original design	State the timeline wherever the i* models are used	Documentation change; applied in Tasks 02 and 03
Show evidence that the original defects are gone	Publish a before/after validation comparison	Appendix B added

Sprint review 4 — 22 October 2026

Increment presented (`v1.0`). The integrated strategic model family; the merged operational model; 68 Camunda Forms; the expanded i* SD and SR models; the alignment evaluation with 44 traced requirements; the acceptance evaluation with 18 executed cases; the deployment and execution evidence.

Against the sprint goal. Everything in the goal was delivered. The models deploy and execute; eighteen acceptance cases passed with no incident.

Not completed. Five criteria remain unmet and cannot be demonstrated on this build: RBAC enforcement, audit immutability, management reporting (the monthly timer has not elapsed), real external integration, and the downtime procedure. The enquiry and reporting processes deploy but were not driven by the execution run. Both are recorded in Task 04, section 6.

Feedback received and the change it produced. Place the operational task identifier in the same traceability row as the strategic node (applied). Give a recommendation against each reported failure rather than only reporting it (applied). Record unimplemented actions with their reason and consequence (applied: two actions remain unimplemented and are logged below).

3. Retrospectives

Sprint 3. *Went well:* the strategic models were produced quickly once the abstraction level was agreed; the second-owner rule held. *Did not go well:* the SR model slipped, and the i* assumptions note was written in fragments and contradicted itself once. *Improvements carried into Sprint 4:* agree the abstraction level in writing before modelling; write the assumptions note in one sitting; start the acceptance review earlier.

Sprint 4. *Went well:* deploying the models to a real Camunda 8 cluster was the single most valuable verification step of the project — it found three XSD sequence violations and a runtime form-binding defect that the moddle round-trip, the lint engine and the Modeler had all passed. Reporting those four defects honestly and then closing them with a clean 18-case run was more defensible than reporting a test plan that had only ever run against stubs. *Did not go well:* the sprint was planned at 80 hours against 48 nominal, and the final integration landed in the last two days. *On a further release:* rehearse integration at mid-sprint rather than at the end; build the test harness before the exception paths are specified; treat documentation as scheduled work.

4. Records of the marked standups

SU3 — 15 October 2026, 09:30, all four present

Member	Completed	Next actions	Blockers and handover
max	S3-02: S1_Strategic_Landscape.bpmn with 24 nodes and 7 external participants; abstraction note drafted	Finish S3-03 (six-lane referral-to-authorisation model) and start the strategic-to-operational map	None blocking. Needs finn's actor list before finalising external participant naming. henry holds the landscape file and can take over.
finn	Draft SD model S3-09: 12 actors and 27 dependencies rendered from the JSON model source; i* tooling decision closed	Start S3-10 (SR decompositions) and draft the soft goals	The renderer does not yet draw the resource-dependency arrowhead correctly, so the SVG export is held. henry is second owner on S3-10.
henry	S3-13 requirements list closed at 44 identifiers with source sentences; S3-15 filename convention and DoD added to the README; S3-08 version snapshots; traceability started at 35 of 44 rows	Finish the nine open traceability rows, then draft the alignment narrative	The open rows are the non-functional requirements (RBAC, audit, availability, reporting) and cannot be closed until the model boundary is agreed. max is second owner on the traceability table.
peter	S3-14 revised criteria table with original wording retained; S3-16 repeatable lint and audit export	Run the lint audit against the three strategic files and prepare the SR3 acceptance slide	One Camunda 8 error remains: a message boundary event without a subscription correlation key. Proposed as a Sprint 4 item rather than patched mid-sprint.

SU4 — 21 October 2026, 09:30, all four present

Member	Completed	Next actions	Blockers and handover
max	S4-05 integration into the three-model family; S4-06	Finish the abstraction note and produce the exports	The exports need peter's lint result before they are declared

Member	Completed	Next actions	Blockers and handover
	message-flow register; S4-07 correlation-key fix		final. henry is second owner on the abstraction note.
finn	S4-12 SD v1.0 (16 actors, 40 dependencies); S4-14 soft goals with conflicts recorded; S4-15 assumptions reconciled	Finish the SR coverage and co-sign the SD/SR consistency checklist	Dependency: the alignment evaluation needs the final SD element identifiers. henry is waiting on those.
henry	S4-16 traceability complete: 44 identifiers, of which 28 fully supported, 12 partly, 2 not supported and 2 out of scope, each with a strategic activity and an implementation artefact; S4-17 ten prioritised improvements; S4-21 portfolio assembly started	Complete the alignment narrative, then run the hand-in check	Assembly needs the frozen i* PDFs and the final exports. max is second owner on the traceability table; peter holds the acceptance evidence.
peter	S4-08 deployment and re-validation; S4-19 execution of the acceptance cases	Complete the criteria quality check and the change log, then the hand-in check	Blocker resolved: deployment found and fixed three XSD sequence violations and the form-binding defect. finn is second owner on the test evidence.

5. Feedback action log

ID	Point raised	Agreed response	Owner (1st / 2nd)	Status
FB-01	SR2: the booking model does not show an unavailable external service	Add the pending-booking path with a recorded retry	max / finn	Implemented (T_KeepPending)
FB-02	SR2: acceptance criteria must be measurable	Rewrite with a measurement method; retain the original wording	peter / henry	Implemented
FB-03	SR2: the repository has no review rule	Protect the main branch and record contribution evidence	henry / max	Implemented
FB-04	SR2: presentation claims must be supported by the case study	Add a source column to the assumptions and re-check the requirement list	henry / peter	Implemented
FB-05	SR3: the SR model must be defensible branch by branch	Complete the SR coverage; model the unsupported urgency branch as a soft goal rather than a rule	finn / henry	Implemented
FB-06	SR3: prove the criteria are measurable	Add a measurement method and a coverage map	peter / finn	Implemented
FB-07	SR3: pool names do not match the process names	Rebuild as one model family with a single participant register	max / henry	Implemented

ID	Point raised	Agreed response	Owner (1st / 2nd)	Status
FB-08	SR3: show evidence that the original defects are gone	Publish a before/after validation and deployment comparison	peter / max	Implemented (Appendix B)
FB-09	SR3: the i* work must not be presented as informing the original design	State the timeline wherever the i* models are used	finn / henry	Implemented
FB-10	SR3: agree the urgency classification rules	Obtain the rule from the module team and model the escalation	henry / finn	**Not implemented.** The hospital has not agreed a complete set of urgency rules (case study, enquiries paragraph) and the module team confirmed none would be issued before hand-in. *Implication:* urgency remains discretionary, so the urgent-enquiry path is a soft goal rather than a rule, and an urgency-rules decision is a second-release Must.
FB-11	SR3: add automated RBAC regression tests	Automate the role-access and audit checks	peter / finn	**Not implemented.** Sprint 4 had no capacity once the deployment defects and both evaluations were re-scoped. *Implication:* RBAC and audit immutability are evidenced by design inspection only, and a regression suite is a second-release Should.
FB-12	SR4: show the operational task identifier in the traceability row	Combine the strategic and operational columns	henry / max	Implemented
FB-13	SR4: give a recommendation against each reported failure	Add a recommendation per failure	peter / finn	Implemented
FB-14	SR4: unimplemented actions are acceptable if the reason is recorded	Record the reason and the outstanding implication	henry / all	Implemented (see FB-10, FB-11)

6. Definition of Done

A task is Done when all of the following hold:

- It meets the acceptance conditions agreed at sprint planning.
- A second team member has reviewed it.
- It is in the shared repository.
- It is linked to its evidence.
- Models parse with zero moddle warnings and lint with zero Camunda 8 reports.

Exports are regenerated from the same revision as the model.

The task record shows status, current progress, the next action and any handover.

Applied to the final increment: the four models satisfy conditions 1 to 6, and the two evaluations satisfy 1 to 4 and 7.

7. Contribution matrix

Member	BPMN models	i* models	Requirements, traceability and evaluations	Agile, documentation and repository	Standups and reviews	Total
max	45%	10%	10%	10%	25%	**25%**
finn	15%	60%	10%	10%	10%	**25%**
henry	15%	15%	45%	60%	35%	**25%**
peter	25%	15%	35%	20%	30%	**25%**

Each of the five activity columns shows the share of that activity's work carried out by each member, and each column sums to 100%. The Total column is the split of the group mark, which the group agreed at the start of Sprint 1 to divide equally: **25% each**. The two are not the same measure — the activity columns record who carried which work, the Total column records how the group agreed to share the mark — which is why a member can hold a large share of one activity and still take an equal share overall.

The distribution follows the roles agreed at the start of Sprint 1 rather than being assigned after the fact.

henry led the two activities that carry the most weight in the assessment. On **requirements, traceability and evaluations** he owned the requirements list and user stories (PB-05, PB-06), the requirements traceability matrix (PB-07), the alignment evaluation and its 44-requirement trace (PB-36, PB-37) and the change log and criteria revisions (PB-38 to PB-40), and he wrote the alignment narrative. On **agile, documentation and repository** he owned the product and sprint backlogs and the definition of done (PB-03, PB-04), the repository, README and filename convention (PB-02), the version snapshots (S3-08, S3-15), the feedback action log (PB-42), the retrospectives (PB-43) and the portfolio assembly and hand-in (PB-44). At the **standups and reviews** he presented the traceability and evaluation findings at both reviews and prepared the records for all four marked activities, which is the largest share of that column.

max retains the largest single-domain share — 45% of the BPMN work, as integration lead — and **finn** 60% of the i* models, as their author. **peter** carries 35% of the evaluation activity through the acceptance testing and the forms, and 30% of the standup and review activity, since he presents the execution evidence at each review.

The equal split was agreed because no member carried the project alone: **max** held the integration, **finn** the i* models and the external workers, **peter** the forms and the testing, **henry** the requirements, evaluations and documentation. The activity columns above show where each of those contributions landed; the Total column is equal by agreement rather than by arithmetic on those columns.

Evidence is the task ownership recorded in the backlog and the contribution record retained in the repository for tutor moderation.

References

Camunda (2025) *Camunda 8 documentation: Zeebe, BPMN, forms and the Orchestration Cluster API*. Available at: https://docs.camunda.io/ (Accessed: 20 October 2026).

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Object Management Group (2013) *Business Process Model and Notation (BPMN), Version 2.0.2*. OMG document formal/2013-12-09. Available at: https://www.omg.org/spec/BPMN/2.0.2/ (Accessed: 20 October 2026).

UWE Bristol (2026) *Case Study: Hospital Patient Referral, Treatment and Administration System*. Unpublished module case study, UFCEP4-0-3 Business Process Modelling and Enterprise Architecture. Bristol: University of the West of England.

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Appendices

Appendix A — Model inventory

Model	File	Process definition(s)	Contents
O1 operational merged model	`O1_Operational_Merged.bpmn`	`PR_Operational_Merged`	8 lanes, 59 flow nodes, 71 sequence flows, 15 message flows, 8 external participants
S1 strategic process landscape	`S1_Strategic_Landscape.bpmn`	`PR_Landscape`	1 pool, 24 flow nodes, 15 message flows, 7 external participants
S2 strategic collaboration	`S2_Strategic_Referral_To_Authorisation.bpmn`	`PR_ReferralToAuthorisation`	6 lanes, 36 flow nodes, 9 message flows, 5 external participants
S3 strategic collaboration	`S3_Strategic_Treatment_To_Aftercare.bpmn`	`PR_TreatmentToAftercare`, `PR_EnquiryHandling`, `PR_ManagementReporting`	7 + 3 + 1 lanes, 74 flow nodes, 13 message flows, 6 external participants

Lane attribution, task identifiers, gateway conditions, form references and Zeebe job types for every model are listed in `08_Repository_Documents/model-inventory.md`, which is generated from the model definitions rather than written by hand.

Appendix B — Validation, deployment and execution

B.1 Model validation. Each model was checked with the toolchain Camunda Modeler itself uses.

Check	Result
Schema round-trip (`bpmn-moddle` 10.1.0, Zeebe 1.18 and modeler extensions)	0 warnings in all four models
Camunda compatibility lint (`@camunda/linting` 3.57.0, Camunda Cloud target)	0 reports in all four models
Structural audit (connectivity, reachability, gateway defaults, duplicate identifiers, empty lanes)	0 defects
Layout audit (nodes inside lanes, no overlaps)	0 defects

B.2 Deployment and execution. The models were deployed to a running Camunda 8 Run 8.10.0-alpha5 cluster through the Orchestration Cluster API.

Check	Result
Process definitions deployed	6 of 6
Camunda Forms deployed	68 of 68, generated from the models
Acceptance test cases executed	18
Cases passed	18
Incidents during the run	0
Defects found by deploying and executing	4 (DEP-01 to DEP-04), all fixed and re-verified

The per-case record — instance key, final state, every element instance reached, the variables set at each step and the completion trace — is in `08_Repository_Documents/test-evidence/acceptance-run.json`, with the console log alongside. Operate and Tasklist screenshots are in `05_Test_Evidence/screenshots/`.

B.3 Defect summary. Sixteen defect categories were found in the Sprint 1 and Sprint 2 working files; fifteen are fixed and one is superseded by the rebuild. Four further defects were found by deploying and executing. The full log, with the correction and the closing evidence for each, is `08_Repository_Documents/defect-log.md`.

Appendix C — Submitted files

Folder	Contents
`02_BPMN_Models/`	The four `.bpmn` models and `forms/`, the 68 Camunda Forms. Deploy the models and the forms together and the processes run end to end
`02_BPMN_Models/readable_exports/`	PDF (single page, zoomable), PNG and SVG for each model, plus detail panels for O1, S2 and S3
`06_iStar_Models/`	SD and SR as PDF, PNG and editable SVG, plus `iStar_models_source.json`, the model source from which both are generated
`07_Portfolio_Documents/`	This document, plus the same content split into sections
`05_Test_Evidence/`	Validation output, the deployment record, the acceptance run JSON and log, Operate and Tasklist screenshots, and the worker-driven scenario runs in `worker-run/`
`04_Java_Worker/`	The Java job workers: one worker per job type, with the Maven build, run scripts, `README.md` and `DEPLOYMENT.md`
`08_Repository_Documents/`	Model inventory, defect log, integration decision, i* assumptions, assumption register, improvement proposals, strategic-to-operational map, version snapshots and test evidence

The team's model-definition generators and working notes are retained in the shared project workspace rather than included in this package; the delivered worker source is in `04_Java_Worker/`.

Appendix D — Files to submit

Required by the specification	Where
Strategic BPMN — `.bpmn` files	`02_BPMN_Models/`
Strategic BPMN — readable PDF exports	`02_BPMN_Models/readable_exports/`
i* models — SD and SR PDF exports	`06_iStar_Models/`
i* models — editable source files	`06_iStar_Models/` (SVG and JSON)
Alignment evaluation with a requirements traceability	`07_Portfolio_Documents/`

Required by the specification	Where
table	
Acceptance evaluation with test results and evidence	`07_Portfolio_Documents/`
Product backlog, sprint backlogs, definition of done, retrospectives	`07_Portfolio_Documents/`
Records of the two marked standups and the two sprint reviews	`07_Portfolio_Documents/`
Feedback action record and contribution evidence	`07_Portfolio_Documents/`
External workers — source, dependencies and configuration	`04_Java_Worker/` (`pom.xml`, `src/`, `README.md`, `DEPLOYMENT.md`)
Worker-driven execution evidence	`05_Test_Evidence/worker-run/`
Repository links	this submission; the repository documents are included as `08_Repository_Documents/`